12-10 第2篇

阅读理解（说明文） Why Does Time "Fly" When We Are Busy?

① Most of us have noticed this small, stubborn fact of life. A long, empty Sunday afternoon, with nothing to do, seems to last forever: we look at the clock, and only **24 minutes** have passed. A busy Saturday, in contrast — a morning dentist visit, a lunch, a train to an event, a quick run to the shop — feels like it lasts **two hours**, not eleven. Same number of minutes, very different **"felt length"** of time. The phenomenon is sometimes called, in everyday English, *"time flies when you are having fun"* — or, more accurately for our purposes, *"time flies when you are busy."* It turns out this is not an illusion. It is a small, measurable working feature of how the human brain builds our sense of time. Scientists have, in the last thirty years, come up with a careful answer that fits in three words: **attention, memory, and prediction.**

② **Piece one: attention and the "time-cell" clock.** Deep in the brain's **hippocampus** (the part that stores memory), there is a population of cells known simply as **time cells**. These cells, discovered in the early 2010s by researchers at Boston University, fire in a careful sequence during any task, creating a kind of internal "clock" of an event. The more attention we pay to time itself ("how long is this lesson going to last?"), the more energy these cells use, and the more time-markers they lay down. Conversely, when we are absorbed in something — drawing, talking, writing, planning a trip — we pay **almost no attention to time passing**. The internal clock still runs, but we are not reading it. A long period can slip by in what feels like ten minutes. **Time, in the felt sense, is a sum of how many "now" labels the brain notices.**

③ **Piece two: memory and the "remembered length" rule.** An important second truth about felt time is that **in the moment**, time that flies feels short; but **in memory**, that same period often feels surprisingly **long**. This is called **the holiday paradox**, first carefully described by psychologist Claudia Hammond in 2012. A rich, busy week-long holiday full of new places, strange foods, and unfamiliar smells feels, on the last evening, like it has been **about two weeks**. Why? Because the brain, when it tries to *remember* a period, estimates length by **the number of different events stored**. A busy week produces hundreds of new memory items. A normal week at school produces fewer. In the moment, busy = fast; in memory, busy = long. This is why **"doing a lot" paradoxically expands our sense of a full life**, even though each busy hour felt short while we lived it.

④ **Piece three: prediction and surprise.** The brain is, deep down, a prediction machine. If the next second is exactly as expected, the brain almost does not register it. Only when a surprise happens — an unexpected sound, a new taste, a person's face at a door — does the brain carefully lay down a "new event marker." A 2021 University of

California study put **120 students** in two rooms for exactly 30 minutes. **Room A**: boring, identical, quiet. **Room B**: small unexpected events (a clock chime, a new smell through the ventilation, a short clip of birds). At the end, Room A students estimated they had spent an average of **42 minutes**; Room B students estimated **26 minutes**. **Boredom**, in the moment, **lengthens** the felt time — because the brain keeps asking "is it over yet?" and time-cells keep firing in attention. **Novelty shortens it** — because the brain is busy with new data. Surprising, isn't it? But it fits the rule.

⑤ **Piece four: why childhood summers felt enormous.** You may have noticed: a week in August when you were seven felt like a small era. A week in December when you are fourteen feels like, well, a week. This is not romantic confusion. It is biology. A child's world is almost entirely made of new events — new words, new objects, new places, new faces. Every hour of a seven-year-old's day produces far more "new markers" than an hour of a forty-year-old. A 2019 meta-study at Duke University estimated that **subjective time between ages 5 and 10 runs about 2—3 times slower** than subjective time after age 40. This is one careful reason why old people say, *"the years go by faster and faster":* after decades of the same routines, new markers are rarer. **To slow down the felt speed of your adult life, introduce novelty.** This is not a slogan. It is testable.

⑥ **What about the opposite: when time seems to stop?** In certain emergencies — a near-miss car accident, a fall — many people report time slowing down to a strange cinematic level: "I could see the cup falling for what felt like three seconds, although it was a tenth of a second." A 2019 Baylor College study measured this carefully by dropping subjects from a safe 30-metre tower onto a net (with consent). Subjects, despite their felt "slow motion," did **not** actually see finer detail than in normal time. The brain, under high stress, lays down memory at a much faster rate — so, when remembered a few seconds later, the event feels long and detailed. It is a memory effect, not a perception effect. Fear, it turns out, is a super-charged version of the attention-and-memory rule.

⑦ **So how do we use this wisely?** The science gives us **three practical rules**. **(1) If today is slow and you want it to be fast**, stop watching the clock; get absorbed in one task. Ten minutes of real work is better than twenty minutes of watching time pass. **(2) If this year feels short and you want it to feel long**, add small novelty: a new route home, a new small hobby, a new subject explored on weekends, a new walk in a new park. These add "event markers" your brain will remember. **(3) If an exam minute feels endless**, you are reading the time cells, not the exam. Look back at your paper; work; the minute will become, surprisingly, almost invisible.

⑧ The **final word**: time is a real physical thing, measured exactly by atomic clocks. But the **time we feel** is an experience, built in the brain out of **attention we pay, memories we store, and surprises we meet**. Two thirteen-year-olds can sit for the same hour; to one it is a lifetime, to the other a whisper. Their hearts beat the same number of times. What differs is the number of **small "now" labels** they noticed. Knowing this, **you can almost, quietly, edit your own experience of time**. On a boring afternoon: become

curious, not restless. On a tiring week: introduce one kind new event. On a life: **pay attention, collect small memories, let yourself be surprised**. In doing so, you grow not only older, but, in the human felt sense, **larger**.

注释： hippocampus 海马体；paradox 悖论；novelty 新鲜感；subjective 主观的；atomic clock 原子钟

根据短文内容，选择最佳答案。

1. According to Paragraph ②, what are "time cells" and how does attention affect them?

A. Tiny blood cells in the eye.
B. A population of brain cells in the hippocampus (discovered by Boston University researchers in the early 2010s) that fire in a careful sequence during a task, creating an internal "clock" of an event; the more attention we pay to time itself, the more energy these cells use and the more time-markers they lay down, while if we are absorbed in a task we pay almost no attention to time passing and a long period can slip by.
C. Small muscles in the ear.
D. Cells that store food energy.

2. What is the "holiday paradox" first carefully described by Claudia Hammond in 2012?

A. Holidays are always short.
B. Holidays are always exactly the same in memory as in the moment.
C. In the moment, busy or exciting time feels short; in memory, however, the same rich period feels surprisingly long, because the brain estimates remembered length by the number of different events stored — so a busy week-long holiday full of new places and smells feels like about two weeks on the last evening.
D. Holidays feel long while they last and short afterwards.

3. What did the 2021 University of California study with 120 students in two rooms find?

A. Both rooms produced the same time estimate.
B. Room A (boring, identical, quiet) students estimated they had spent 42 minutes in a 30-minute session, while Room B (small unexpected events — a clock chime, a new smell, a bird clip) students estimated 26 minutes; boredom lengthens felt time while novelty shortens it.
C. Boredom shortens felt time.
D. Both groups estimated 60 minutes.

4. According to the 2019 Duke University meta-study, what is true about subjective time between ages 5 and 10 compared to after age 40?

A. They are the same.
B. Time between ages 5 and 10 runs about 2—3 times slower than after age 40 — one careful reason childhood summers felt enormous and one reason old people say the years go faster; to slow the felt speed of adult life, introduce novelty.
C. Time after 40 runs slower.
D. Ages 5 and 10 are only 10% slower.

5. What is the main idea of the final paragraph?

A. Time is only a machine-measured quantity; human feeling does not matter.
B. Felt time is built in the brain from attention, memories, and surprises — two thirteen-year-olds can sit for the same hour with very different "felt lengths" depending on the small "now" labels they noticed; knowing this, you can almost quietly edit your own experience of time by being curious rather than restless, introducing novelty, paying attention, collecting small memories and letting yourself be surprised, growing not only older but larger in the human felt sense.
C. Boredom is good for the brain.
D. Only childhood time is meaningful.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段时间细胞
【解析】"海马 / 序列 / 注意→标记"直引。A、C、D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第③段假期悖论
【解析】"当下快 + 记忆长 + 事件数量"直引。A、B、D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第④段加州研究
【解析】"42 vs 26"直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段杜克元研究
【解析】"5-10 岁慢 2-3 倍"直引。A、C、D 均错。故选 B。

5.
【答案】B
【题型】主旨大意题
【定位】第⑧段结尾
【解析】"编辑自己的时间 / 更大"主旨。A、C、D 均偏离。故选 B。